

1. Administrative & Study Identification

Full Study Title: Efficacy and Safety of Switching From Anti-C5 Antibody Treatment to Iptacopan Treatment in Study Participants With Atypical Hemolytic Uremic Syndrome (aHUS) **Official Title:** A Multicenter, Single Arm, Open-label Study to Evaluate Efficacy and Safety of Switching From Anti-C5 Antibody Treatment to Iptacopan Treatment in Study Participants With aHUS **Short Title/ Acronym:** aHUS Switch Study **Protocol Number:** CLNP023F12302 **NCT Number:** NCT05935215 **Posting ID:** 391483665624333882 **Sponsor/ Company Name:** Novartis Pharmaceuticals **Investigational Product:** Iptacopan hydrochloride (Fabhalta®) - Oral factor B inhibitor **Phase of Development:** Phase 3 **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of iptacopan (measured by the absence of thrombotic microangiopathy [TMA] manifestations) upon switching from an anti-C5 antibody treatment to iptacopan treatment in study participants with atypical hemolytic uremic syndrome (aHUS). **Secondary Objectives:** To assess the effects of iptacopan on a range of efficacy assessments relevant to aHUS, including hematological parameters such as the change from baseline in platelet count at month 12 and month 24, and specific TMA manifestation tracking.

2.2 Background & Rationale aHUS is a very rare, progressive, and life-threatening disease characterized by the destruction of red blood cells, decreased platelets, and kidney inflammation leading to renal failure, driven by the uncontrolled activation of the alternative complement pathway (AP). This Phase 3 study evaluates if participants who have achieved clinical stability on intravenous or subcutaneous anti-C5 therapies can safely and effectively transition to oral iptacopan hydrochloride, thereby reducing treatment burden while preventing disease recurrence.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A **Trial Status:** RECRUITING

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 50 participants **Number of Sites:** Approximately 15 countries

worldwide **Estimated Study Start:** 28-Feb-2024 (Actual) **Estimated Primary Completion:** 21-Jul-2028 (Estimated Study Completion: 19-Jul-2029)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female adult participants ≥ 18 years of age with a diagnosis of aHUS for whom etiologies of other types of TMA and non-aHUS kidney disease have been excluded. 2. Currently on the recommended weight-based dosage regimen of anti-C5 antibody treatment for at least 3 months prior to the screening visit. 3. Clinical evidence of response to anti-C5 antibody treatment (in absence of PE/PI) for at least 3 months prior to entering the screening period as defined by: - Hematological normalization in platelet count $\geq 150 \times 10^9/L$ and LDH below upper limit of normal (ULN), and - Stable or improving kidney function as defined by $\leq 15\%$ increase in serum creatinine. 4. Vaccination against *Neisseria meningitidis* and *Streptococcus pneumoniae* infections is required prior to the start of treatment with iptacopan. 5. If not received previously or if a booster is required, vaccination against *Haemophilus influenzae* infection should be given, if available and according to local regulations.

4.2 Exclusion Criteria 1. History of aHUS disease relapse while on anti-C5 antibody treatment. 2. eGFR < 30 ml/min/1.73m². 3. Active infection or history of recurrent invasive infections caused by encapsulated bacteria, i.e., meningococcus, pneumococcus (e.g., *N. meningitidis*, *S. pneumoniae*) or *H. influenzae*. 4. Participants with sepsis or active systemic bacterial, viral (including COVID-19), or fungal infection within 14 days prior to study treatment administration. 5. Kidney, bone marrow transplant (BMT)/hematopoietic stem cell transplant (HSCT), heart, lung, small bowel, pancreas, liver transplantation or any other cell or solid organ transplantation. 6. Female patients who are pregnant or breastfeeding, or intending to conceive during the course of the study. 7. Any medical condition deemed likely to interfere with the patient's participation in the study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Iptacopan hydrochloride 200 mg twice a day (BID). **Route of Administration:** Oral **Duration of Treatment:** Up to 8 weeks screening period, followed by a 12-Month Core Treatment period and a 12-Month Extension Treatment period (24 months total).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for aHUS and management of underlying

comorbidities. **Prohibited:** Concurrent use of anti-C5 antibodies following the successful transition to iptacopan therapy.

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures
Screening	Up to 8 Weeks	Informed Consent, Medical History, Baseline Labs, Confirmation of required vaccines (Meningococcal/Pneumococcal/H. influenzae).
Baseline	Day 0	Treatment Switch: Discontinue anti-C5, First Dose of Iptacopan.
Interim	Months 1 to 12	Efficacy Assessments (Platelets, LDH, Creatinine tracking), Safety & Infection monitoring.
End of Study	Month 24	Final Vital Signs, Exit Interview, IP Accountability, Final TMA occurrence review (End of Extension Period).

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Percentage of participants free of TMA manifestation:** Absence of thrombotic microangiopathy (TMA) manifestation, without use of anti-C5 antibody, during the 12 months of iptacopan treatment following the switch of treatment from an anti-C5 antibody to iptacopan treatment.

7.2 Secondary Outcomes 1. Percentage of participants free of TMA manifestation in study participants with functionally significant mutations in complement genes or positive anti FH antibodies. 2. Percentage of participants free of TMA manifestation. 3. Time to TMA manifestation. 4. Change from baseline in platelets at month 12 and month 24. 5. Sustained hematological normalization and stability of serum creatinine across the core and extension treatment periods.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via HCP assessment, with particular vigilance for signs of encapsulated bacterial infections. **Serious Adverse Events (SAEs):** Protocol-mandated rapid reporting (typically 24 hours), especially concerning meningococcal or pneumococcal infection risk. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee utilizing Novartis standard pharmacovigilance tracking.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy assessment; Safety Set for AE/SAE evaluations. **Sample Size Justification:** \$N=50\$ participants provides adequate power to

assess the safety and sustained efficacy of the switch in this ultra-rare disease population. **Handling of Missing Data:** Last Observation Carried Forward (LOCF) and Multiple Imputation approaches for longitudinal laboratory endpoints (platelets, LDH).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote monitoring to verify protocol adherence, specifically verifying strict vaccination prerequisites. **Audit Trail:** eCRF locking and comprehensive data clarification mechanisms executed by the sponsor.

11. Study Identifiers & Governance

Primary ID: CLNP023F12302 **Registry Number:** NCT05935215 **Posting ID:** 391483665624333882 **Sponsor/Lead Organization:** Novartis **Study Title:** Efficacy and Safety of Switching From Anti-C5 Antibody Treatment to Iptacopan Treatment in Study Participants With Atypical Hemolytic Uremic Syndrome (aHUS) **Official Regulatory Title:** A Multicenter, Single Arm, Open-label Study to Evaluate Efficacy and Safety of Switching From Anti-C5 Antibody Treatment to Iptacopan Treatment in Study Participants With aHUS

12. Clinical Framework (aHUS Specific)

Disease Area / Condition: Atypical Hemolytic Uremic Syndrome (aHUS) **Preferred UMLS Name:** Therapy for atypical hemolytic uremic syndrome **MeSH Code:** D065766 **SNOMED ID:** 789660001 **Diagnosis Threshold:** Clinically stable on anti-C5 therapy (Platelets $\geq 150 \times 10^9/L$, LDH < ULN, stable or improving serum creatinine) **Medical Methodology:** Alternative Complement Pathway Factor B inhibition **Optimization Target:** Sustained remission of TMA manifestations on oral monotherapy

13. Study Procedures & Methodology

Management Pathway: Transition from maintenance IV/SC anti-C5 antibodies to oral iptacopan hydrochloride **Education Protocol (HCP):** Strict verification of meningococcal, pneumococcal, and *H. influenzae* vaccinations prior to administration **Education Protocol (Patient):** REMS (Risk Evaluation and Mitigation Strategy) counseling on identifying signs/symptoms of severe bacterial infections. **Quality Audit Mechanism:** Routine laboratory panels and clinical symptom tracking to detect any early signs of relapse or TMA.

14. Observation & Follow-Up Schedule

Total Duration: 24 treatment/follow-up weeks per segment (12-Month Core + 12-Month Extension) **Onsite Visit Cadence:** Baseline, interim milestone months (e.g., Months 3, 6, 12, 18, 24) **Remote Monitoring Frequency:** Standard protocol-defined check-ins **Checklist Review Interval:** Continuous monitoring of infection risks and hematological markers

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]					
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]					